

Variables:-

A variable is like a container(box) that we use to store data.

```
let name = "vaishvik";
let age = 24;
let isTrue = true;
let location = {
  city : "shimla",
  state : "Himcahal"
};
let skills = ["Java", "JS", "React.js", "MySQL", "MongoDB", "Node.js"];
```

How to name your variable:-

PascalCase :- UserName (React.js)(Not used in JavaScript)

kebab-case :- user-name (html attributes, css)

snake_case :- user_name (used in java and other programming languages)(not used in JS)

camelCase :- userName (used in JS when naming variables, functions and other features of JS)

Variables special cases:-

- variable name can't start with a number.
- variable can't start with special characters.
- Reserved keyword should not be used to name a variable for eg:- let , const

Variables are of three types :-

- 1) var :- original way of declaring a variable. Not used anymore in modern JavaScript. But still used in legacy code.
- 2) let :- can change value later.(introduced in es6, in 2015)
- 3) const :- cannot change value later. (introduced in es6, in 2015)

Re-assign a variable

Example of re-assignment:-

```
let a = 10;  
a = 20; // this is re-assignment (perfectly fine cause let is used and can be  
re-assigned)
```

```
const b = 20;  
b = 30; // would throw an error because const can't be re-assigned
```

```
var c = 30;  
c = 40; // would be re-assigned because var is used.
```

- 1) variables declared with let can be re-assigned
- 2) variables declared with const can't be re-assigned
- 3) variables declared with var can be re-assigned

Re-declare a variable

Example of re-declaration

```
let a = 10;  
let a = 100; // would throw an error because let can't be re-declared
```

```
const b = 10;  
const b = 100; // would throw an error because const can't be re-declared
```

```
var c = 10;  
var c = 100; // would not throw an error because var can be re-declared
```

Scope of variables :-

```
let is blocked scoped  
const is blocked scoped  
var is function scoped
```